

Background

- Oral oncolytics are a high-risk, complex class of medications that are often challenging to manage in outpatient settings and pose increased risks in hospitalized patients.
- A medication use evaluation was conducted at Saint Luke's Health System over 6 months (October 2020 - April 2021), reviewing 51 inpatient oral chemotherapy orders.
 - 14 orders were deemed potentially inappropriate due to clinical instability, organ dysfunction, or admission for treatment-related toxicity. All were initiated by non-oncology providers.
 - 6 significant drug interactions were noted, with incomplete documentation of Hematology/Oncology consultation.

Objective

- This study aimed to compare the effectiveness of a best practice alert (BPA), initiated by the Saint Luke's Cancer Institute and Clinical Decision Support Team in October 2023, designed to alert non-oncology providers, nurses, and pharmacists when an oral oncolytic is ordered or reordered from a prior to admission medication list.

This intervention aims to enhance safety by ensuring appropriate use and timely consultation.

Primary Endpoint

- Incidence of potentially inappropriate oral oncolytic orders during inpatient admission.

Study Design/Methods

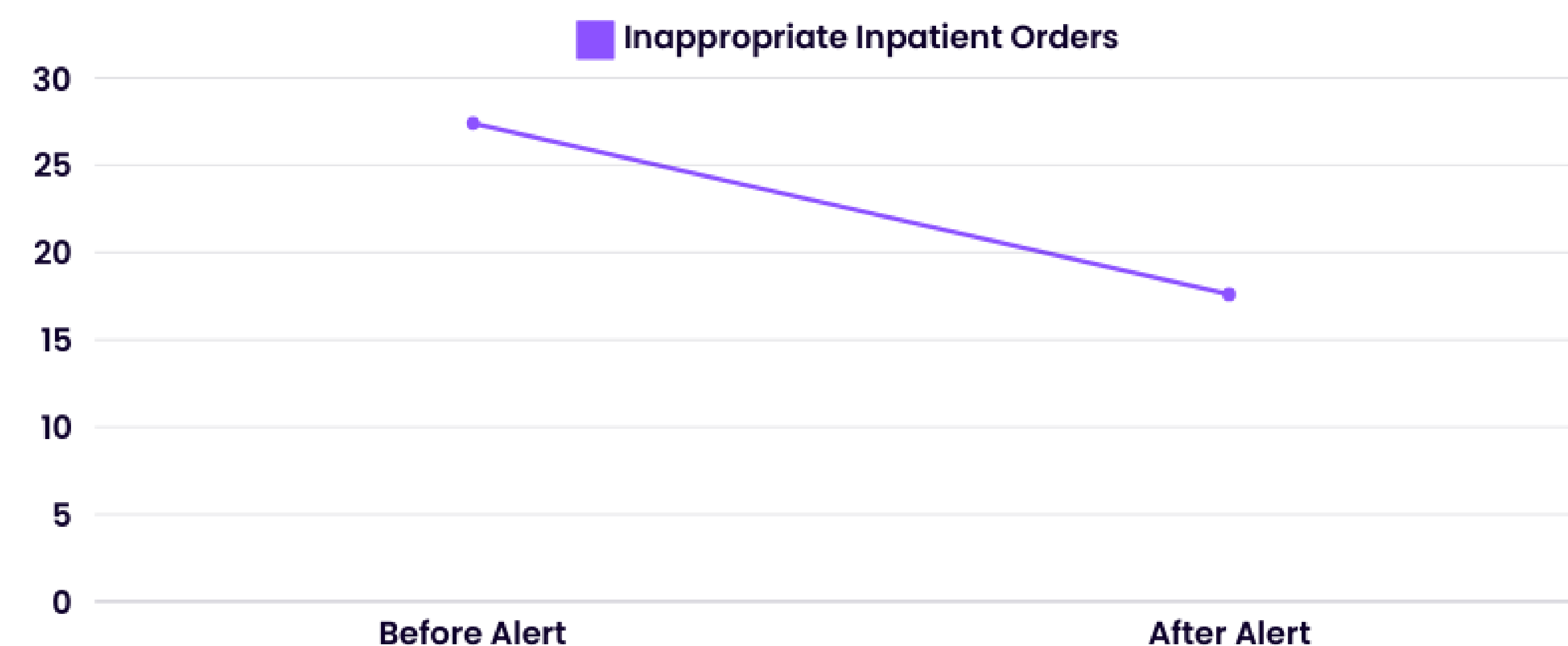
- This retrospective, single health system chart review evaluated patients admitted to Saint Luke's Health System hospitals between November 28, 2023, and August 18, 2024, who were receiving oral chemotherapy prior to admission.

Study Design/Methods

- The BPA requires the provider to justify the order, with options to indicate consultation with Hematology/Oncology.
- Methotrexate, hydroxyurea, and hormonal therapies were excluded to reduce alert fatigue due to their common use in non-oncology settings.

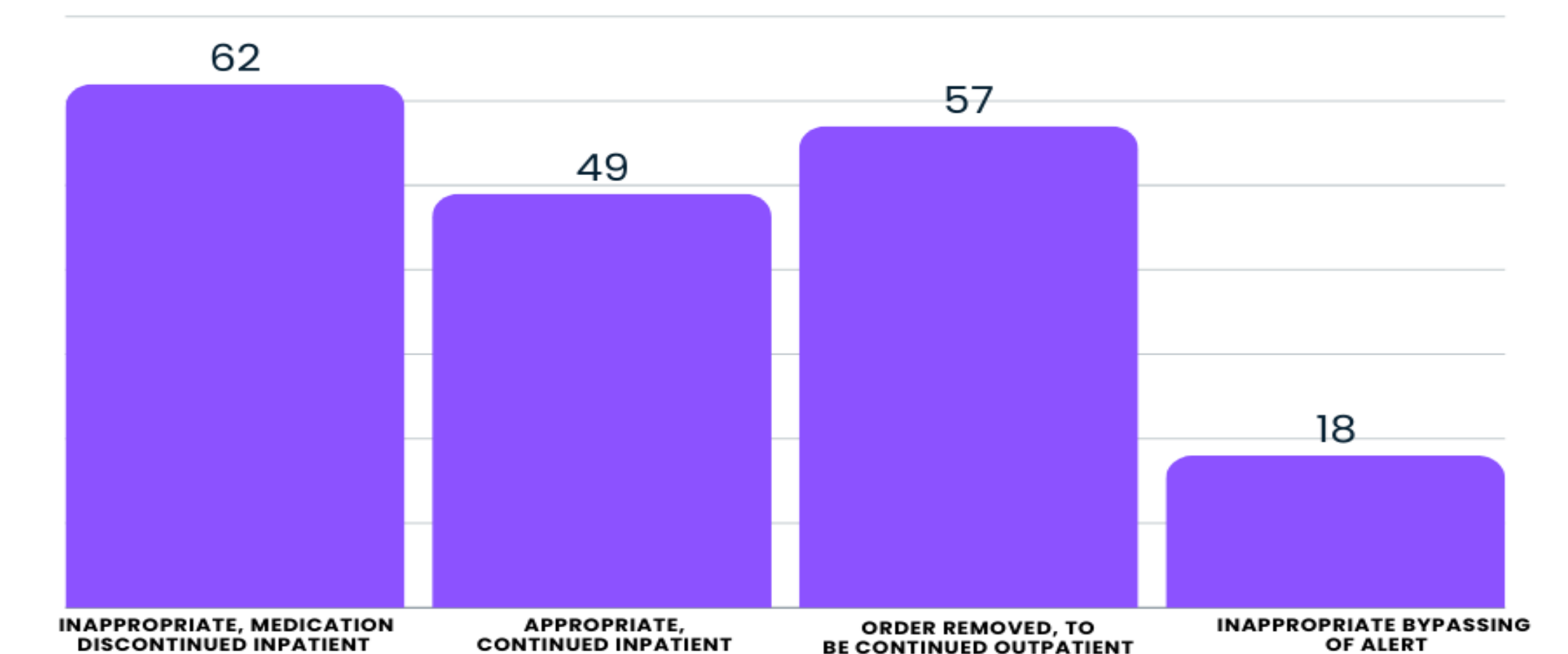
Data Collected
Admission Date
Patient Demographics
Oral oncolytic regimen
Provide/service line
Inpatient Administration (Yes/No)
Best Practice Alert (BPA) outcomes: Appropriate (medication continued), Inappropriate (medication discontinued), Inappropriate BPA bypass

Results



Results

- Of 279 BPAs triggered, 186 were analyzed after excluding duplicates, errors, or non-oncology indications.
- Prior to BPA implementation, 14 out of 51 (27.45%) oral oncolytic orders were considered inappropriate, with at least one dose administered during hospitalization. Post-BPA, only 9 out of 51 (17.6%) inappropriate orders were administered.



Conclusion

- The reduction in inappropriate oral oncolytic use highlights the BPA's effectiveness in improving inpatient medication management.
- Oral oncolytics carry significant toxicity risks, especially in acutely ill patients with comorbidities. Their complex pharmacokinetics and interactions with common inpatient medications, such as antibiotics and anticoagulants, increase the risk of harmful drug-drug interactions and may reduce treatment efficacy or cause adverse effects.
- Temporarily discontinuing inappropriate oncolytics during hospitalization is crucial to prevent further complications, manage acute conditions, and enhance patient safety by minimizing morbidity and mortality. Careful review and pausing of these therapies optimize overall inpatient care.

Disclosures

The authors have nothing to disclose concerning possible financial or personal relationships with commercial entities that may have a direct or indirect interest in the subject matter.

References

- Smith, M. A., & Johnson, R. T. (2018). 2018 Hematology/Oncology Pharmacist Association Best Practices for the Management of Oral Oncolytic Therapy: Pharmacy Practice Standard. Journal of Oncology Practice, 14(6), 393-400. <https://doi.org/10.1200/JOP.18.00581>